



JSPM's Jayawant Institute of Management Studies, Pune.

Mini Project Synopsis

**Project Title:
Retail Data Engineering Pipeline for
Point-of-Sale Analytics and Incentive Calculation**



**Submitted by,
Ashish P. Mali (Roll No: MCA2531)**

**Project Guide,
Prof. Darshana Surwase**



JSPM's

JAYAWANT INSTITUTE OF MANAGEMENT STUDIES

(NAAC Accredited- B Grade)

(Approved by AICTE, New Delhi, Recognised by Govt. of Maharashtra &
Affiliated to Savitribai Phule Pune University)

S. No. 80, Pune-Mumbai Bypass Highway, Tathawade, Pune - 411033.

Phone: 8237076937/38

E-mail: director_jims@jspm.edu.in **Website:** www.jspmjims.edu.in



Prof. Dr. T. J. Sawant

B.E. (Elec.) PGDM, Ph. D

Founder- Secretary

AISHE ID: C-60096

Dr. Shailesh Rajans

MBA, Ph.D.

Director

Date: / / 2026

Certificate

This is to certify that **Mr. Ashish Popat Mali** is a Bonafide student at Jayawant Institute of Management Studies, Tathawade, Pune has successfully completed the Mini Project (MP581MP) as prescribed by the **Savitribai Phule Pune University** in the partial fulfilment of the requirement of Master of Computer Applications (MCA) Program for the academic year 2026-2027.

Seat No.: _____

Prof. Darshana Surwase
Project Guide

Prof. Leena Deshmukh
H.O.D

Dr. Shailesh Rajhans
Director

Project Synopsis: Retail Data Engineering Pipeline for Point-of-Sale Analytics and Incentive Calculation

1. Proposal for Mini-Project

This project proposes the design and implementation of a scalable data engineering pipeline that ingests Point-of-Sale (POS) retail transaction data, processes and models it into analytics-ready tables (star schema), and produces business reports including store/product KPIs and an automated employee incentive calculation module.

2. Title

Retail Data Engineering Pipeline for Point-of-Sale Analytics and Employee Incentives

3. Problem Statement

Retail organizations collect high volumes of POS and master data across stores, products, and employees, but this raw data is often fragmented, inconsistently typed, and not designed for fast analytics.

Management needs timely KPIs (total sales, sales by store/product, top employees) and a reproducible incentive calculation mechanism to drive sales.

The project addresses these gaps by building a reliable ETL/ELT pipeline using Apache Spark and cloud object storage, producing clean fact/dimension tables and derived reports for business consumption.

4. Objectives

The primary objectives of this project are:

1. Build an ingestion layer that reliably loads raw POS, product, store and employee data into a data lake.
2. Implement Spark-based cleaning, type-correction, deduplication, and transformation jobs.
3. Model data using a star schema (fact_sales and supporting dimension tables) for fast OLAP queries.
4. Produce summary tables and dashboards for KPIs (total sales, revenue by product/store, daily/monthly trends).
5. Implement and validate a rule-based employee incentive calculation module (monthly/quarterly).
6. Demonstrate the pipeline end-to-end (ingest → bronze/silver/gold layers)

→ analytics outputs).

7. Document design decisions, code, and test cases to make the pipeline reproducible.

5. Scope of the Study

The scope includes a prototype implementation and documentation demonstrating:

- Raw data ingestion into an S3-style data lake (or local object store emulator).
- Batch Spark ETL jobs that implement bronze/silver/gold layers (raw → cleaned → analytics).
- Dimensional modelling: one central fact_sales table plus dimensions (product, store, employee, date).
- Production-style features: partitioning, schema enforcement, incremental loads, basic monitoring/logging.
- Analytics outputs: sales_summary, product_performance, store_performance, and employee_incentives.

Excluded: real delivery logistics, full production scheduling/orchestration (Airflow can be suggested but not fully deployed in prototype), or company-wide RBAC.

6. Hardware Requirements

- **CPU:** Intel Core i5 / AMD Ryzen 5 or better.
- **RAM:** Minimum 8 GB (16 GB recommended for faster local Spark runs).
- **Storage:** Minimum 20 GB free disk for project, datasets, and Spark local data.
- **Network:** Stable internet to download dependencies and (optional) connect to cloud services.

7. Software Requirements

- OS: Windows 10 / Ubuntu 20.04+ / macOS (recent).
- Java JDK 11+ (required by Spark).
- Python 3.8+ with packages: pyspark, pandas, boto3, sqlalchemy, psycopg2 (or other DB connector), pyarrow.
- Apache Spark 3.x (local or cluster mode).
- Hadoop client jars (optional if using HDFS).

- Amazon S3 (or local S3 emulator like MinIO) for data lake storage.
 - PostgreSQL / DuckDB / Amazon Redshift (one of these as analytics/serving layer).
 - Jupyter Notebook / PyCharm / VSCode for development.
 - Git for version control; AWS CLI for interacting with S3 if using AWS.
 - (Optional) Docker for running Spark locally or MinIO.
-

8. Modules

The project is structured into six distinct modules:

Module 1: Data Ingestion & Landing (Raw layer)

- Read raw CSV/JSON POS files from local/stream/S3; landing area (bronze).
- Key artifacts: raw file layout spec, ingestion scripts, timestamp bookkeeping.

Module 2: Cleaning & Standardization (Silver layer)

- Type corrections, null handling, deduplication, currency/amount normalization, datetime standardization.
- Key artifacts: cleaned Parquet/Delta tables, data quality checks.

Module 3: Transformation & Enrichment (Gold layer)

- Join sales with product/store/employee dimension data; calculate derived metrics (total_sale, discount applied, taxes).
- Partitioning strategy, incremental load logic.
- Key artifacts: fact_sales (analytics-ready), dimension tables.

Module 4: Dimensional Modelling & Analytics

- Design star schema: fact_sales, dim_product, dim_store, dim_employee, dim_date.
- Build aggregate tables for dashboards (daily_sales, monthly_store_summary).
- Key artifacts: ER diagram, SQL queries for KPIs.

Module 5: Incentive Calculation & Reporting

- Implement rule-based incentive logic (tiered % based on monthly sales) as SQL or Spark UDF.

- Produce employee_incentives table with calculation breakdowns and auditability.
- Key artifacts: incentive rules, sample payout reports.

Module 6: Packaging, Documentation, and Demo

- Unit/integration tests, README, runbook (how to run pipeline end-to-end), sample dataset, and demo notebook.
- Key artifacts: project repo, demo slides/notebook, run scripts.

9. References

- **Apache Spark Official Documentation**
<https://spark.apache.org/documentation.html>
- **Amazon Web Services Documentation (Amazon S3)**
<https://docs.aws.amazon.com>
- **Python Official Documentation**
<https://docs.python.org>
- **YouTube Tutorials on Data Engineering and Apache Spark**